Distributed ledger stack: This layer consists of the core distributed ledger technology stack. Blockchain building, transaction execution and consensus happen in this layer. The components in this stack might vary with the distributed ledger product chosen. However, the common sub-components will be consensus algorithms, data storage, transactions management and other relevant services.

Security identity and access management: This meso-layer handles single sign-on, authentication, encryption, and authorisation capabilities.

IT application and infrastructure management: This meso-layer illustrates the advantages of Infrastructure-as-a-Service and Platform-as-a-Service, which require large computing capabilities.

Types of blockchains

Blockchains are classified as public, private or hybrid depending on the nature of the application. Public and private blockchains have many similarities as well as differences in their functionality.

Public blockchain: This blockchain is completely open to the public, which means anyone will be able to join the network as a participant. A public blockchain typically uses a mechanism to incentivize participating parties, which encourages a growing number of participants in the network. A public blockchain is open for everyone to read, send transactions and participate in the consensus process. The most prominent examples of public blockchain are Bitcoin and Ethereum.

A public blockchain is open source and no one is in charge. There is no access or rights management done for a public blockchain and anyone can be a part of the consensus process. Because of this, anyone at any given point of time can join or leave/read/write/audit the public blockchain ecosystem, and the network will still be trustless.

Table 1: A comparison of public and private blockchains

Functionality	Public	Private
Access	Open read/write	Permissioned read and/or write
Speed	Slower	Faster
Security	Proof of work Proof of stake Other consensus mechanisms	Pre-approved participants
Identity	Anonymous	Known identities
Asset	Native asset	Any asset
Run	Anyone can run BTC/LTC full node	Anyone can't run a full node
Transaction	Anyone can make transactions	Anyone can't make transactions
Audit	Anyone can review/audit the blockchain	Anyone can't review/audit the blockchain
Participants	Permissionless (Anonymous)	Permissioned (Identified, trusted)
Consensus	Proof of work, Proof of stake	Voting or multi-party consensus algorithm
Transaction approval frequency	Long (10 minutes or more)	Short (100x msec)

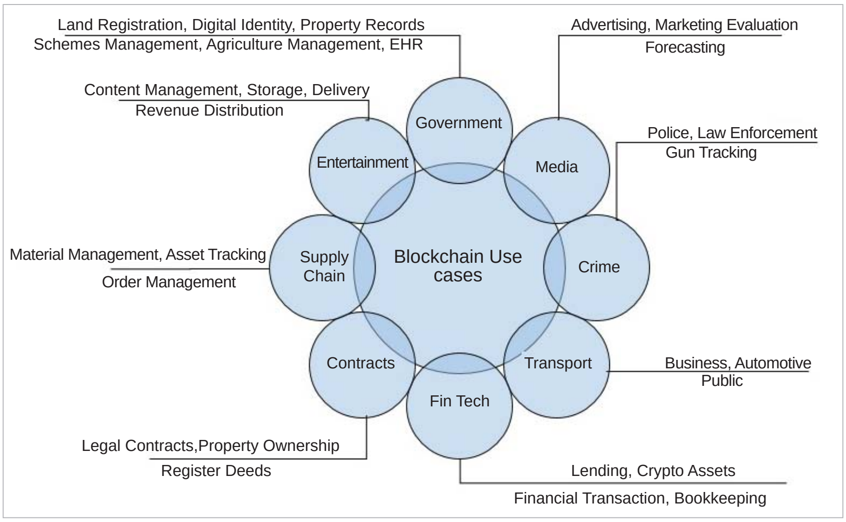

Figure 2: Blockchain use cases

Private blockchain: Private blockchain is an absolute opposite of public blockchain. The access to a private blockchain is limited to the parties involved in the creation of that network, or those granted access to it by the parties who created it. The internal mechanics of a private blockchain can vary — from existing participants serving as administrators who decide on the inclusion of future entrants to simple observers. But the public cannot access the private blockchain.

In private blockchains, the owner of the blockchain is a single entity or an enterprise, which can override/delete